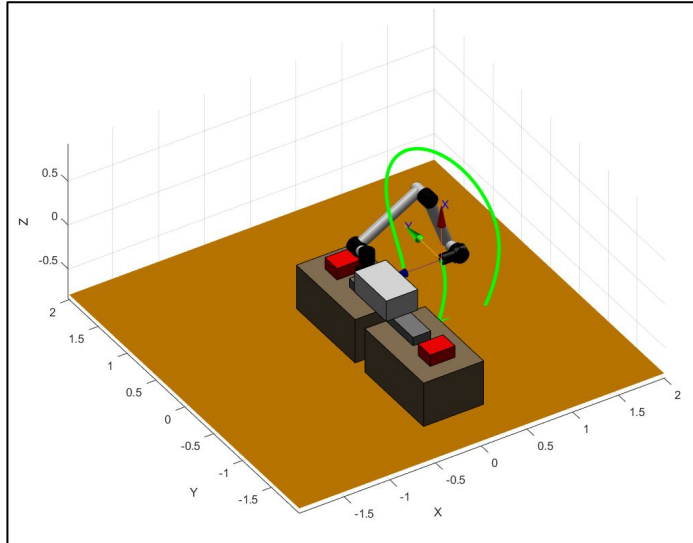


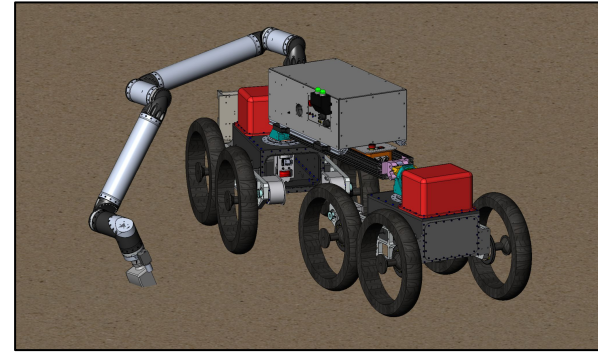
Erik Kramer | Mobile ISAM Robotic Arm Testbed (SAMMI)

Project → An untethered reconfigurable robotic testbed for in-space servicing assembly and manufacturing development

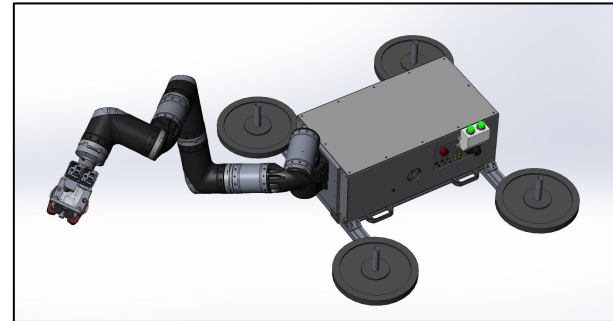
Tasks → Defining requirements, **arm design** and **analysis**, avionics **housing** design, **control software** development



Kinematic Analysis Driven Arm Design



Modular Design Approach



Erik Kramer | Mobile ISAM Robotic Arm Testbed (SAMMI)

Objectives

- Build an ISAM robotic arm testbed capable of digging and manipulation when stable or on a mobile base
- Design a new digging arm that can effectively collect soil while mounted atop a robotic rover base
- Create a base to house avionics and mechanically support different arms

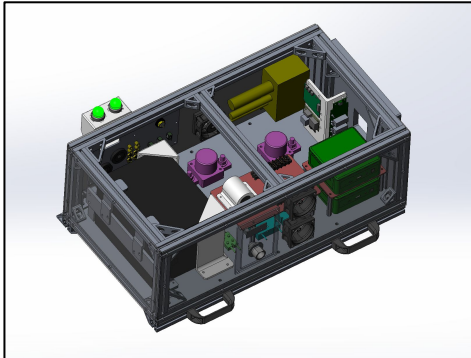
Process

- Utilized **MATLAB** to model arm candidates and select a design
- Employed **SolidWorks** to design new components to interface with existing heritage hardware
- Designed **housing** for electronics and deployed in-house software for **arm control**

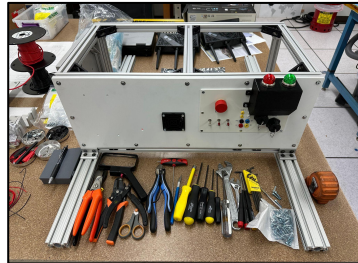
Results

- Produced two physical testbeds that can be configured with different arm types and can be used as a stand alone system or on a mobile base
- Utilized Testbed in ISAM demos including satellite servicing, segmented telescope assembly, and lunar solar panel assembly

Harness Routing Aware Avionics Configuration



Electromechanical Build



System Bring-up and Testbed Operation

